

KARUSH SURI

[Homepage](#) ◇ [Email](#) ◇ [GitHub](#) ◇ [Google Scholar](#) ◇ [LinkedIn](#) ◇ [X](#)

AUTODIDACTIC LEARNING

Reading

Topology, James Munkres, all chapters	2026
Topics in Algebra, I.N. Herstein, all chapters	2025
Optimization Algorithms on Matrix Manifolds, P. Absil, chapters 2-7	2021
Convex Optimization, Stephen Boyd, chapters 6-11	2021
Neuro-dynamic Programming, Dimitri Bertsekas and John Tsitsiklis, chapters 1-6	2021
Abstract Dynamic Programming, Dimitri Bertsekas, chapters 1-2	2020
Machine Learning: A Probabilistic Perspective, Kevin Murphy, chapters 20-24	2020
Pattern Recognition and Machine Learning, Christopher Bishop, chapters 8-12	2019

Writing

Paper Summaries	(link)
Research Work	(link)

Code Implementations

Toy Examples (JAX and PyTorch)	(link)
Hierarchical DQN (JAX)	(link)
Proximal Policy Optimization (JAX)	(link)
Conservative Q Learning (JAX)	(link)
Discrete Gumbel Samplers (JAX)	(link)
Eligibility Traces with Neural Networks (PyTorch)	(link)
Evolution Strategies (PyTorch)	(link)
Lagrangian Algorithms and Vector Products (Autograd)	(link)

EDUCATION

University of Toronto	2025 - 2028 (Expected)
<i>PhD, Electrical & Computer Engineering</i>	<i>Toronto, Canada</i>
Thesis: Tail End Gumbel Post-Training	
GPA: 3.8 / 4 (minor in mathematics)	
University of Toronto	2019 - 2021
<i>MASc, Electrical & Computer Engineering</i>	<i>Toronto, Canada</i>
Thesis: Deep Hierarchical Reinforcement Learning	
GPA: 4/4	
Amity University	2015 - 2019
<i>BTech, Electrical & Computer Engineering</i>	<i>Delhi, India</i>
Thesis: Sign Language Translation from Wearable Sensors (demo)	
GPA: 8.78/10 (rank: 4/142)	

EMPLOYMENT

Valence Labs (Mila)	2023 - 2025
<i>Research Engineer</i>	<i>Montreal, Canada</i>
Advisors: Dr. Emmanuel Bengio	
Projects: (1) Multi-Modal Phenomic Foundational Models, (2) Amortized Inference with GFlowNets	
Google X	2021 - 2023
<i>AI Resident</i>	<i>Mountain View, USA</i>
Advisors: Grace Brentano & Dr. Lam Nguyen	
Project: Undisclosed	

INTERNSHIPS

Lossfunk

Research Intern

Advisors: Paras Chopra

Projects: Understanding Drug Repositioning via Chemical Reasoning

2026 - 2026

Toronto, Canada

Fidelity Investments

Machine Learning Intern

Advisors: Kyle Burgers

Projects: Sparse Transformers for Tabular Quality Control

2025 - 2026

Toronto, Canada

Borealis AI

Student Researcher

Advisors: Xiao Qi Shi

Projects: Deep Hierarchical Reinforcement Learning for Trade Execution

2020 - 2021

Toronto, Canada

PUBLICATIONS

* = equal contribution

"Understanding Judge Calibration in Multi-Turn Debates"

Under Review

Karush Suri, Konstantinos Plataniotis, Yuri Lawryshyn

"A Cross Modal Knowledge Distillation & Data Augmentation Recipe for Improving Transcriptomics Representations through Morphological Features"

ICML 2025

Ihab Bendif, Yassir El Mesbahi, Alisandra Kaye Denton, Karush Suri,

Kian Kenyon-Dean, Auguste Genovesio, Emmanuel Noutahi

"Understanding Conditional Computation in Contrastive Phenomic Retrieval"

Technical Report 2025

Karush Suri, Puria Moghadam, Frederik Wenkel, Maciej Sypetkowski,

Emmanuel Bengio, Emmanuel Noutahi, Dominique Beaini

"How Molecules Impact Cells: Unlocking Contrastive PhenoMolecular Retrieval"

NeurIPS 2024

Philip Fradkin*, Puria Moghadam*, Karush Suri, Frederik Wenkel,

Ali Bashashati, Maciej Sypetkowski,

Dominique Beaini

FM4S @ NeurIPS 2024 (oral)

(best paper award)

"On the Scalability of GNNs for Molecular Graphs"

NeurIPS 2024

Maciej Sypetkowski, Frederik Wenkel, Farimah Poursafaei,

Nia Dickson, Karush Suri, Philip Fradkin, Dominique Beaini

"Surprise Minimizing Multi-Agent Learning with Energy-based Models"

NeurIPS 2022

Karush Suri, Xiao Qi Shi, Konstantinos Plataniotis, Yuri Lawryshyn

"Off-Policy Evolutionary Reinforcement Learning with Maximum Mutations"

AAMAS 2022 (oral)

Karush Suri

"Continuous Sign Language Recognition from Wearable IMUs using Deep CapsNet and Game Theory"

CEE, Elsevier, 2019

Karush Suri, Rinki Gupta

"Transfer Learning for sEMG-based Hand Gestures using Deep Learning in a Master-Slave Architecture"

IEEE IC3I 2018

Karush Suri, Rinki Gupta

PENDING PATENTS

"Generating Actions for a Supply Chain Network"

2024

Lam Nguyen, Grace Brentano, Sze Lee, Karush Suri, Anikait Singh,

Salil Pradhan, David Andre

Google X, Application Number: 52862.

"Large Language Model Derived Environment State Changes In Supply Chain Logistics"

2023

Lam Nguyen, Grace Brentano, Salil Pradhan, David Andre, Gearoid Murphy,

Sze Lee, Karush Suri, Raja Panjwani, Anikait Singh, Klara Kaleb

Google X, Application Number: 52750.

“ <i>Large Language Model Interface for Supply Chain Networks</i> ” Lam Nguyen, Grace Brentano, David Andre, Salil Pradhan, Anikait Singh, <u>Karush Suri</u> Google X, Application Number: 52503.	2023
“ <i>Generating Network Alignment Information</i> ” Raja Panjwani, Anikait Singh, Ashish Chona, Sze Lee, Grace Brentano, <u>Karush Suri</u> , Lam Nguyen, Salil Pradhan Google X, Application Number: 52766.	2022

SERVICES

Reviewer, <i>TMLR</i>	2026 - Present
Reviewer, <i>ICML</i>	2024 - 2026
Reviewer, <i>ICLR</i>	2024 - 2026
Reviewer, <i>NeurIPS</i>	2024 - 2026
Teaching Assistant, <i>Digital Image Processing</i> , University of Toronto	2021
Teaching Assistant, <i>Computational Thinking</i> , University of Toronto	2021
Teaching Assistant, <i>Computer Networks</i> , University of Toronto	2020
Teaching Assistant, <i>Computer Organization</i> , University of Toronto	2020

AWARDS

Academic

NSERC Canada Graduate Research Scholarship	2026 - 2029
University of Toronto Doctoral Scholarship	2025 - 2029
Hatch Graduate Scholarship	2025 - 2026
Laura Bassi Scholarship (declined)	2025
Electrical & Computer Engineering Fellowship, University of Toronto	2020 - 2021
Edward S. Rogers Graduate Scholarship, University of Toronto	2019 - 2020
Best in Technical Innovation Award (Class of 2015-2019), Amity University	2019
Most Frugal Innovation Award, Amity University	2018
100% Curriculum Merit Scholarship, Amity University	2015
Young Achievers in Mathematics Award	2015

Industrial

Outstanding Reviewer, NeurIPS, ICML	2023, 2026
Outstanding Intern Buddy, Google X	2022